

COMP4107 Assignment 3

Due: March 6th, 2026

Question 4: Experiments

Group 111

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A)

Findings:

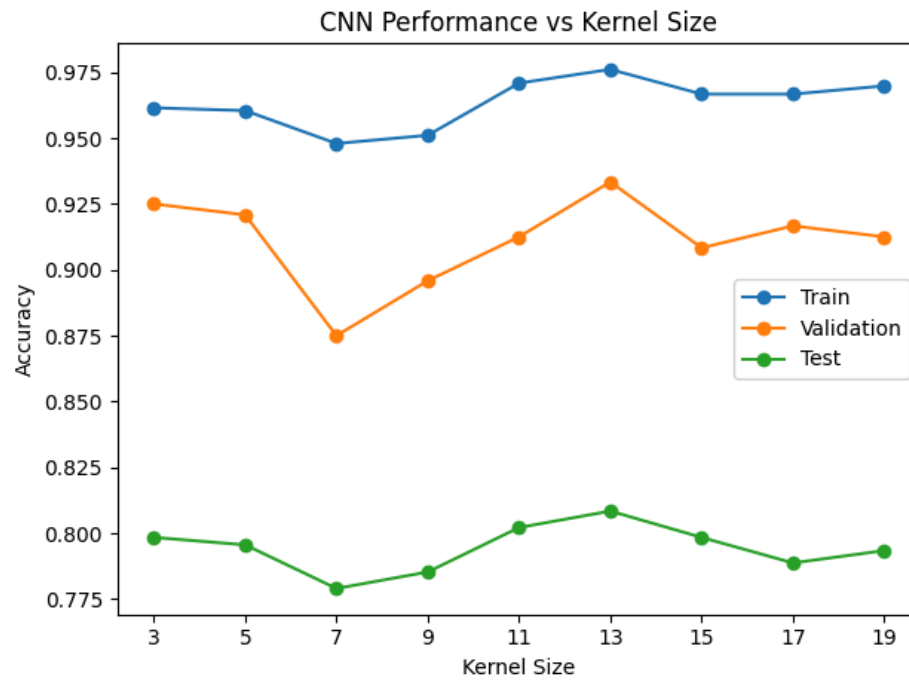


Figure 1: CNN Performance vs Kernel Size

Each kernel size was evaluated across 10 independent training runs of 40 epochs, with the reported accuracies representing the mean over all runs. Training accuracy remains stable between 95-98% across all kernel sizes. Validation accuracy is more volatile (88-93%), which can be attributable to a small validation set size where a single misclassification has a significant effect on total accuracy. Test accuracy is consistently lower (78-81%), which is expected as the test set is fully unseen data. Kernel size 7 appears to be a weak spot across all three curves, performing worse than other values, while kernel sizes 11-13 appear to be more accurate. The persistent gap between train, validation, and test accuracy suggests mild overfitting, though it remains relatively stable across kernel sizes indicating this is not strongly influenced by kernel size alone.

B) Experiments on the size of the hidden states in the recurrent neural network:



Figure 2: RNN Performance vs Hidden States

Hidden state sizes of size 8, 16, 32, 64, 128, and 256 were evaluated independently and trained for 50 epochs, with the reported accuracies representing the mean over all runs. We see an upward trend with training data accuracy with a slight decrease in accuracy with a hidden state size of 128. The training accuracies range from 60-100% on hidden state sizes greater than 8. Validation accuracy is more stable (70%-80%). Test accuracy is consistently lower (78-81%), which is expected as the test set is unseen data.

C)

Findings:

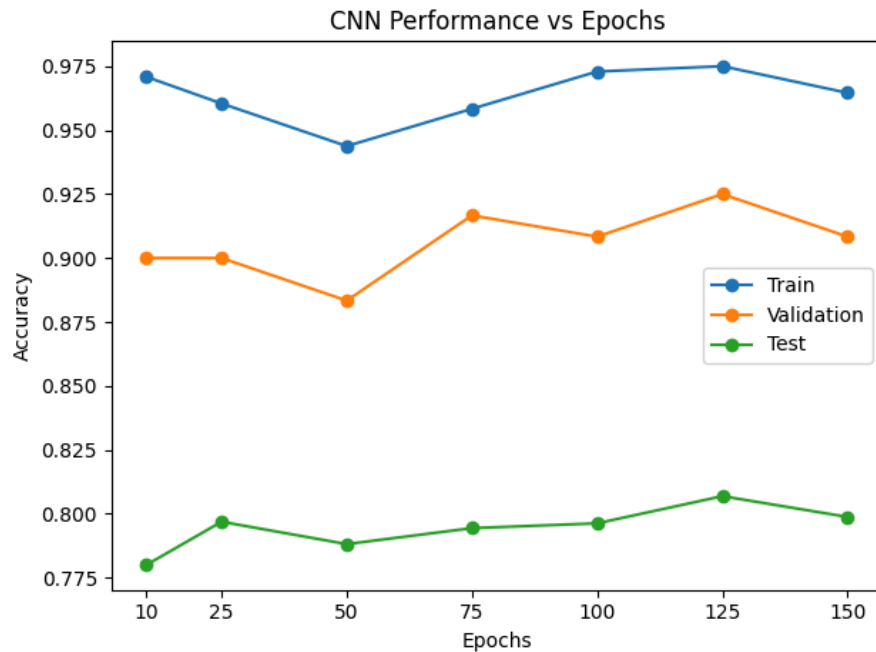


Figure 3: CNN Performance vs Epochs

Each epoch count was evaluated across 5 independent training runs, with the reported accuracies representing the mean over all runs. Training accuracy remains consistently high (95-98%) across all epoch counts, indicating the model converges quickly and maintains its fit on the training data. Validation and test accuracy are more volatile due to small dataset sizes, but both show a modest upward trend peaking around 125 epochs before slightly declining at 150, suggesting the onset of overfitting. The gap between train and test accuracy (~17%) remains consistent across all epoch counts, indicating that slight overfitting is present regardless of training duration. Based on these results, 125 epochs appears to be the optimal stopping point for this CNN with kernel size 13.

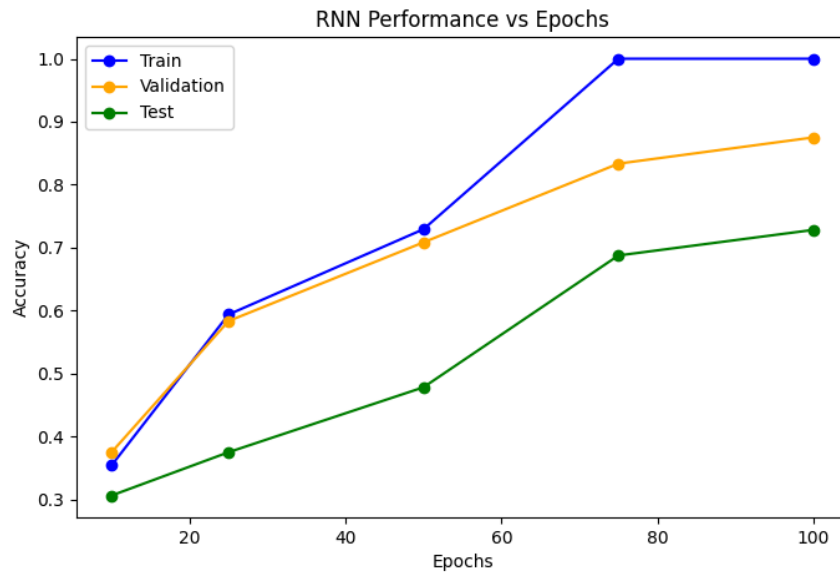


Figure 4: RNN Performance vs Epochs

Number of epochs of 10, 25, 50, 75, and 100 were evaluated and trained independently, with the reported accuracies representing the mean over all runs. We see an upward trend with all three datasets (training, validation, and test). The training accuracy approaches 100%, validation accuracy approaches 90%, and test accuracy approaches 70% as the number of epochs increases. Ultimately, the best performing model is the one trained with 100 epochs.

D)

As seen in Figure 1, Figure 2, Figure 3, and Figure 4 above, the best performing models against the hold out test set are as follows:

CNN:

- Best kernel size is 13 with accuracy: ~81%
- Best number of epochs is 125 epochs with accuracy: ~82%

RNN:

- Best number of hidden states is 256 with accuracy: ~69%
- Best number of epochs is 100 with accuracy: ~73%